INCH-POUND

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SUPERSEDING
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PERFORMANCE SPECIFICATION SHEET

# ELECTRON TUBE, POWER
# TYPE 8295A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Pentode, external anode, ceramic-metal.

See figure 1.

Mounting position: Any.

Weight: 3 pounds (1.36 kg) nominal.

ABSOLUTE RATINGS:

| Parameter: | F | Ef | Eb | Ec1 | Ec2 | Ec3 | Ib |
|---|---|---|---|---|---|---|---|
| Unit: | MHz | V ac | V dc | V dc | V dc | V dc | mA dc |
| Class AB1 (rf or audio) | | | | | | | |
| Maximum: | 30 | 6.3 | 3,000 | --- | 600 | 100 | 800 |
| Minimum: | --- | 5.7 | --- | --- | --- | --- | --- |
| Test conditions: | --- | 6.0 | 2,500 | Adj | 500 | 0 | 400 |

ABSOLUTE RATINGS:

| Parameter: | Ic1 | Pp | Pg1 | Pg2 | tk | Anode core and seal temp | Cooling |
|---|---|---|---|---|---|---|---|
| Unit: | mA dc | W | W | W | sec | °C | 1/ |
| Class AB1 (rf or audio) | | | | | | | |
| Maximum: | 0 | 1,000 | 0 | 30 | --- | 250 | --- |
| Minimum: | --- | --- | --- | --- | 180 | --- | --- |
| Test conditions: | --- | --- | --- | --- | 300 | --- | 2/ |

GENERAL:

Qualification: Not required.



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TABLE I. Testing and inspection.

| Inspection | Method MIL-STD-1311 | Notes | Conditions | Acceptance Level 6/ | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|---|
| Conformance inspection, part 1 | | | | | | | | |
| Pulsing emission (sinusoid) | 1231 | --- | prr = 10 (min); eb = ec2 = ec1 = 1,000 v | 0.65 | is | 70 | --- | a |
| Electrode voltage (grid) | 1261 | --- | | 0.65 | Ec1 | -85.0 | -115.0 | V dc |
| Total grid current | 1266 | 5/ | | 0.65 | Ic1 | --- | -50 | µA dc |
| Electrode current (screen) | 1256 | --- | | 0.65 | Ic2 | --- | 15 | mA dc |
| Conformance inspection, part 2 | | | | | | | | |
| Heater current | 1301 | --- | | --- | If | 7.7 | 8.7 | A ac |
| Direct-interelectrode capacitance (ground cathode connection) | 1331 | --- | | --- | Cgp<br>Cin<br>Cout | ---<br>36.0<br>16.5 | 0.12<br>44.0<br>20.5 | pF<br>pF<br>pF |
| Amplification factor | 1316 | --- | g1 to g2 amplification; Ec1/Ic2 = 60 mA dc; Eb = 0 | --- | Mu | 3.0 | 3.8 | --- |
| Low-frequency vibration | 1031 | --- | No voltages applied | --- | --- | --- | --- | --- |
| Low-frequency vibration end-points: | --- | | | | | | | |
| Electrode voltage (grid) | 1261 | --- | | --- | Ec1 | -85.0 | -115.0 | V dc |
| Total grid current | 1266 | --- | | --- | Ic1 | --- | -50 | µA dc |
| Current division (short pulse, method B) | 1372 | --- | Eb = Ec2 = 500 V dc<br>Ec1 = -200 V dc;<br>egk/ib = 2.5 a | --- | egk<br>ic2 | ---<br>--- | 0<br>400 | v<br>ma |

See footnotes at end of table.



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TABLE I. Testing and inspection - Continued.

| Inspection | Method MIL-STD-1311 | Notes | Conditions | Acceptance level 6/ | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Min | Max | |
| Conformance inspection, part 3 | | | | | | | | |
| Stability | --- | 3/ | Eb = 3,000 V dc; Ec1/Ibo = 220 mA dc at t = 0; Read ΔIb at t = 180 sec | --- | ΔIb<br>t | --- | 22 | mA dc |
| Linear amplifier power output and distortion | 2204 | 3/ | Class AB1 amplifier; Eb = 3,000 V dc; Ec1/Ibo = 220 mA dc; Eg1/Ib = 800 mA dc (1-tone); Rℓ = 2,000 ± 100 ohms; Rg = 1,000 ohms (max); anode tank Q = 10 to 15 | --- | Po<br>3rd IM<br>5th IM | 1.350<br>-20<br>-30 | ---<br>---<br>--- | W<br>dB<br>dB |
| Life test | --- | 4/ | Group C; linear amplifier Power output and distortion except, Eg1/Po = 1,400 W (min) (1-tone); t = 500 hours | --- | --- | --- | --- | --- |
| Life-test end points: | 2204 | --- | | | | | | |
| Linear amplifier power output and distortion | | | | --- | Po<br>3rd IM<br>5th IM | 1,250<br>-19<br>-29 | ---<br>---<br>--- | W<br>dB<br>dB |

1/ With an anode dissipation of 1,000 watts and an incoming air temperature of 50°C at sea level, a minimum of 24 cfm of air shall be passed through the anode cooler. In all applications, an air-system socket, such as the EIMAC SK-184 or SK-184A, or equal, should be used. With an airflow of 24 cfm, the static pressure drop for the tube and socket is approximately 0.13 inch of water. Cooling air should be applied before or simultaneously with the application of electrode voltages, including the heater, and may be removed simultaneously with them. In cases where long life and consistent performance are factors, cooling in excess of minimum requirements is normally beneficial.

2/ During all electrical tests involving application of heater voltage, the use of an air-system socket is permissible, with forced-air cooling of 24 cfm maximum for air at 25°C at sea level. Correction may be made for higher altitude or temperature.

3/ The following tests, listed under conformance inspection, part 3, shall be performed every 3 months, with sample size and acceptance on zero defect criteria as follows:

n1 = 4 c1 = 0

In case of a failure on a stability-test sample, the lot shall be reprocessed and a new sample submitted. the linear amplifier power output and distortion test shall be performed on the sample that is used for the stability test. In case of a failure of a linear amplifier power output and distortion test sample, this test shall become a conformance inspection, part 2, with an acceptance level of 6.5 (see 6/). After three consecutive successful submissions, the test may revert to the conformance inspection, part 3 test.

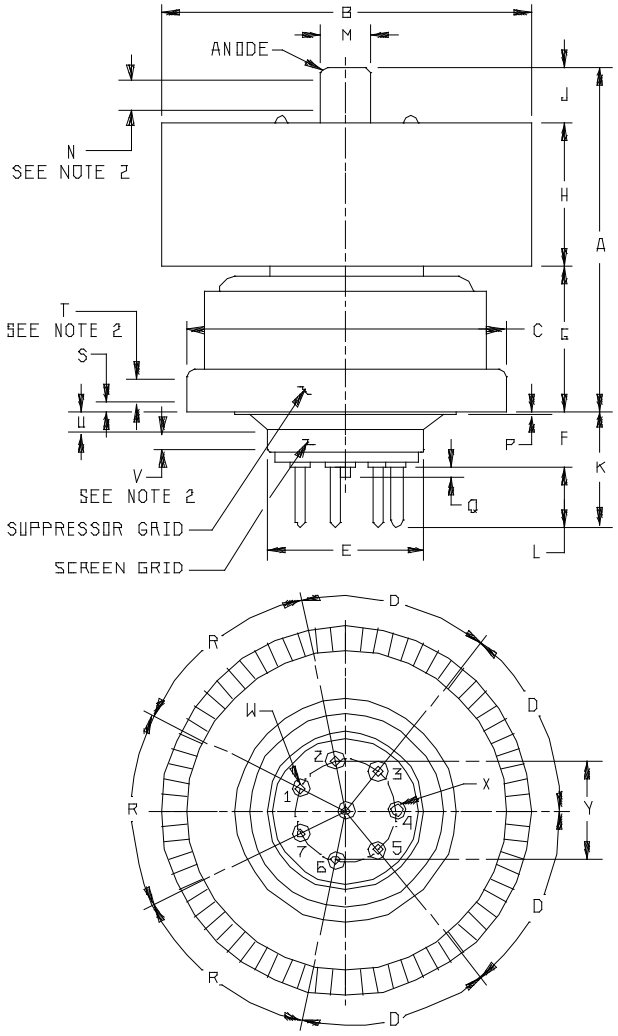
4/ This is a destructive test. All voltage and current meters shall be calibrated for ± 2 percent or better accuracy as installed in the equipment. Prior to the performance of the life test, the tube shall have met the requirements of the linear amplifier power output and distortion test.

5/ This test is to be the first test performed at the conclusion of the holding period.

6/ This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.



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| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| Conformance inspection, part 2 | | | | |
| A | 3.458 | 3.832 | 87.83 | 97.33 |
| C | 3.485 | 3.515 | 88.52 | 89.28 |
| E | 1.615 | 1.630 | 41.02 | 41.40 |
| F | .655 | .719 | 16.64 | 18.26 |
| K | 1.056 | 1.219 | 26.82 | 30.96 |
| M | .559 | .573 | 14.20 | 14.55 |
| W | .056 | .062 | 1.42 | 1.57 |
| X | .120 | .127 | 3.05 | 3.23 |
| Conformance inspection, part 3 (see note 1) | | | | |
| B | 3.968 | 4.032 | 100.79 | 102.41 |
| G | 1.395 | 1.645 | 35.43 | 41.78 |
| H | 1.468 | 1.532 | 37.29 | 38.91 |
| J | .593 | .657 | 15.06 | 16.69 |
| L | .438 | .562 | 11.13 | 14.27 |
| N | .400 | --- | 10.16 | -- |
| P | --- | .125 | --- | 3.18 |
| T | .250 | --- | 6.35 | --- |
| V | .220 | --- | 5.59 | --- |
| Reference dimensions | | | | |
| D | 51° | | 51° | |
| Q | .125 | | 3.18 | |
| R | 52° | | 52° | |
| S | .125 | | 3.18 | |
| U | .250 | | 6.35 | |
| Y | 1.000 | | 25.40 | |

| PIN CONNECTIONS | |
|---|---|
| Pin No. | Element |
| 1 | k |
| 2 | g1 |
| 3 | h |
| 4 | k |
| 5 | h |
| 6 | g1 |
| 7 | k |
| Center pin | int con |
| Lower ring | g2 |
| Upper ring | g3 |
| Cap | a |

NOTES:
1. Dimensions shall be checked yearly. An accept on zero defect sampling plan shall be used, with sample of three tubes with an acceptance number of zero. In the event of failure, the test will be made as a part of conformance inspection, part 2, acceptance level of 6.5 (see 6/). The yearly sampling plan may be reinstated after three consecutive samples have been accepted.
2. Contact surface shall be confined to this area.

FIGURE 1. Outline drawing of electron tube type 8295A.



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Referenced documents. In addition to MIL-PRF-1, this specification sheet references MIL-STD-1311.

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2010-008)

Review activities:
Army - CR4
Navy - AS, CG, MC, OS
Air Force - 99

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